FixMoney: Analytics, Siri Quick-Add aur AI Coins

System design (Hinglish) - Fix-Money-KMP ke liye. Sab free-tier resources pe, scale hone layak.

Item	Detail
Date	6 October 2026
Project	~/Developer/Fix-Money-KMP (Kotlin Multiplatform + Compose; Android + iOS)
Status	**Sirf design.** Abhi koi code nahi likha gaya. Approval ke baad phase-wise implement hoga.
Reference	Old app ~/Downloads/Expense (AdManager.kt, UMP consent, GOOGLE_ADS_ADMOB_INTEGRATION.md), KMP worker/src/index.ts, AddExpenseIntent.swift, CLAUDE.md
Scope	1) Firebase Analytics - end-to-end user journey 2) Siri se bina lamba sentence bole expense add 3) Settings me Siri toggle + permission + one-time "How to use" video (in-app) 4) AI Coins: 5 din free trial, phir rewarded video = 5 coins, 1 coin = 1 AI use

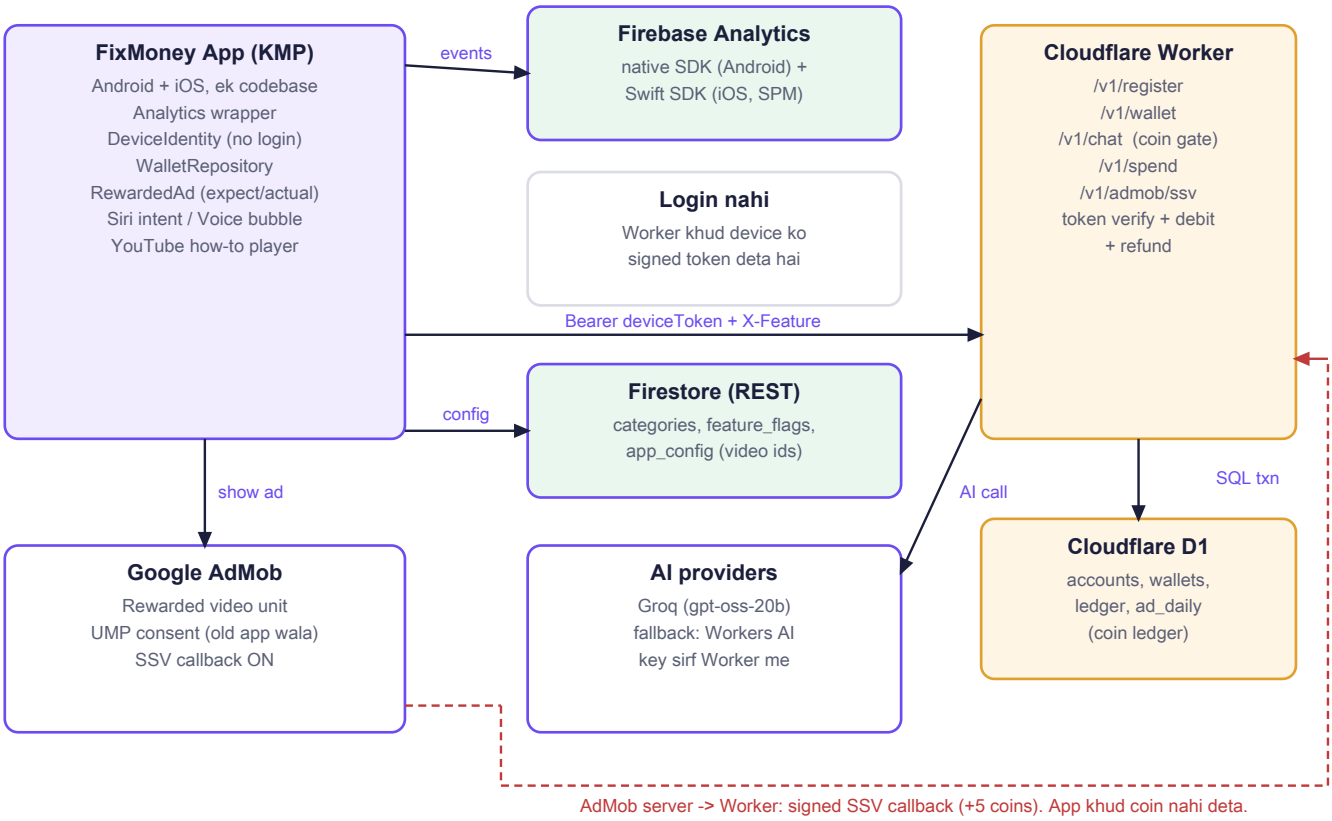
Ek line me: Analytics ke liye native Firebase SDK ek chhote bridge ke through; Siri ke liye chhota phrase + Back Tap / Action Button shortcut (bina "Hey Siri"); coins ka hisaab **server (Cloudflare Worker) pe** hoga, app pe nahi - kyunki saari AI calls pehle se Worker se hi jaati hain. Coins sirf AdMob ke **signed server-side callback** se milenge, isliye koi app hack karke coins nahi bana sakta. Monthly kharcha: Rs 0 jab tak Groq ka free quota kaafi hai.

1. Aaj system kaisa hai (starting point)

Design existing code ke upar bana hai, isliye pehle dekhte hain kya already hai aur kya nahi:

Cheez	Abhi KMP app me	Isse design pe kya asar
AI calls (Fixoo, voice, receipt scan)	Sab **Cloudflare Worker** se jaati hain (fixmoney-ai-proxy.../v1/chat). Groq key sirf Worker me. Groq fail ho to Workers AI fallback.	Coin gate ke liye **perfect jagah**: har AI call already isi se guzarti hai. Bas yahan check + debit add karna hai.
AI Insight card	**AI nahi hai** - HomeInsightEngine local arithmetic hai (3-week baseline, 20% rule). Koi API call nahi.	Isko coin pe rakhna product decision hai, cost ka nahi. Section 7.7 dekhein.
Firebase	Koi Firebase SDK nahi. Categories + feature_flags Firestore **REST** se padhte hain.	Wajah: Firebase ka KMP build nahi hai, aur iosX64 + Kotlin 2.2.0 klib wali do shartein. Isliye bridge approach (section 3).
Ads	KMP me koi ads nahi.	Old app ka AdManager + UMP consent port karna hoga, aur **Rewarded** naya add karna hoga.
Old app ads	Banner + Interstitial live, UMP consent, App ID ca-app-pub-7178541936408370~9464292553. **Rewarded unit nahi hai.**	AdMob console me naya "Rewarded - AI coins" unit + SSV callback banana hoga.
Old app analytics	firebase-analytics dependency hai, lekin **ek bhi custom logEvent nahi** - sirf auto events.	Journey tracking zero se design karni hai. Same Firebase project (expenseapp-7b500) use hoga, history continuous rahegi.
iOS Siri	AddExpenseIntent: "Hey Siri, add an expense to FixMoney" -> Siri poochta hai -> Groq parse -> Room save. Simulator pe probe se verified; real Siri hop kabhi test nahi hua.	Isi intent ko chhota/fast banana hai, toggle se gate karna hai.
Apple Developer account	Nahi hai (TEAM_ID khaali).	Siri entitlement, App Store, real-device Siri test, iOS AdMob - sab iske bina block. Section 10.

2. Poora architecture – ek nazar me



3 golden rules jo poore design me follow hote hain:

- **Paisa/coins ka hisaab sirf server pe.** App sirf balance dikhata hai. App ko root/patch karke koi coin nahi bana sakta, kyunki debit aur credit dono Worker + D1 me hote hain.
- **Coins sirf AdMob ke signed callback se.** App ka "ad dekh liya" message trust nahi hota - AdMob ka server Worker ko ECDSA-signed request bhejta hai, Worker signature verify karke hi +5 deta hai.
- **Expense tracking hamesha free.** Manual add, budget, analytics, export - kisi pe coin nahi. Coin sirf un features pe jo hamare liye AI quota kharch karte hain. Ye AdMob policy ke liye bhi zaroori hai (core app ko ad ke peeche lock nahi karna).

3. Feature 1 – Firebase Analytics (end-to-end user journey)

3.1 Approach: native SDK + chhota bridge

Firebase ka koi official KMP SDK nahi hai, aur community wrapper ko wahi do shartein paar karni padti hain jo is project me har library pe lagti hain (iosX64 + Kotlin 2.2.0 klib). Isliye **Kotlin Firebase ko link hi nahi karega**. Shared code sirf ek interface jaanta hai; asli SDK har platform ka apna native hota hai:

Layer	Kya	Kahan
Common	interface AnalyticsSink { log(name, params); setUserProperty(k, v); setUserId(id) } + object Analytics (typed events)	shared/analytics/
Android	com.google.firebase:firebase-analytics (BOM) - FirebaseAnalytics.logEvent()	composeApp androidMain
iOS	FirebaseAnalytics Swift Package Xcode me (iosApp). Swift class Kotlin ka AnalyticsSink protocol implement karti hai aur launch pe register hoti hai.	iosApp/ + composeApp iosMain
Siri cold start	Siri intent bina UI ke app process chalata hai - FirebaseApp.configure() App ke init me, MainViewController ke bharose nahi (same reason jaise ensureKoinStarted).	iOSApp.swift

> **Kyun ye best hai:** iOS SDK Xcode link karta hai, Kotlin/Native nahi - isliye klib/iosX64 wali problem lagti hi nahi. Interface composeApp me rakhna hoga (shared ki declarations ComposeApp framework header me export nahi hoti - CLAUDE.md me already noted).

3.2 Setup steps (one-time, free)

- Firebase console (expenseapp-7b500) -> Android app already registered hai (com.vikram641.expensetracker). Debug build ka package **.kmp** suffix wala hai - use alag Android app ke roop me add karo, warna debug events aayenge hi nahi.
- iOS app register karo: bundle id **com.vikram641.fixmoney** -> GoogleService-Info.plist iosApp me.
- google-services Gradle plugin composeApp me (Android only).
- Debug: Android `adb shell setprop debug.firebase.analytics.app <pkg>`, iOS scheme me `-FIRDebugEnabled` -> Firebase DebugView me live events.
- Optional: GA4 -> BigQuery export (sandbox free, 10 GB) - raw events pe SQL funnel chahiye ho tab.

3.3 Event taxonomy (journey order me)

Naming rule: snake_case, max 40 chars, har event max 25 params. **Kabhi bhi** note text, naam, ya exact amount nahi bhejna - amount sirf bucket me (0-100, 100-500, 500-2k, 2k+).

Journey stage	Event	Key params
Install / open	first_open, session_start, app_open (auto)	-
Screen tracking	screen_view (manual - Compose me Activity change nahi hota; NavHost destination listener + tab change se)	screen_name, screen_class
Onboarding	onboarding_step_view, onboarding_complete, onboarding_skip_back	step (welcome/feature_1..3/name/budget/currency/all_set), currency, budget_bucket, categories_count
Core tracking	expense_add, expense_edit, expense_delete	**source** (manual/voice_bubble/siri/scan), category, amount_bucket
Budget	budget_set, budget_remove, budget_alert_shown	category, scope (month), percent_bucket
Analytics / Home	month_change, analytics_scope_change, insight_view	screen, scope
Fixoo AI	fixoo_open, fixoo_message_sent, fixoo_reply, fixoo_new_chat	input (typed/suggestion), result (ok/error/no_coins), tier (groq/workers_ai), latency_bucket
Voice (Android)	voice_toggle, voice_permission_result, voice_capture	permission (overlay/mic), granted, result (saved/parse_fail/no_speech/no_coins)
Siri (iOS)	siri_toggle, siri_permission_result, siri_how_to_shown, siri_how_to_play, siri_intent_run	trigger (first_enable/permission_off), percent_watched, result
Receipt scan	receipt_scan_start, receipt_scan_result	stage (capture/review/scan), result (ok/blurry/not_receipt/no_coins)
Coins	trial_started, trial_ended, wallet_open, no_coins_shown, reward_ad_request, reward_ad_shown, reward_ad_completed, coins_credited, coins_spent	feature, ad_result (loaded/failed/closed_early), coins_balance_bucket
Others	export_data, sip_calculate, theme_toggle, settings_toggle, profile_edit	format, result, key, enabled

3.4 User properties (segments ke liye)

Property	Values	Kis kaam ka
currency	INR, USD...	Country/region segment
trial_state	trial / expired / never	Trial ke andar vs baad ka behaviour compare
coin_bucket	0, 1-5, 6-20, 20+	Kitne log coins pe atke hain
voice_enabled / siri_enabled	true / false	Feature adoption
has_budget	true / false	Budget set karne walon ki retention
legacy_user	true / false	Old app se update hokar aaye (LegacyImport chala) ya fresh install

3.5 Dashboards / funnels (GA4 Explorations me)

- **Activation:** first_open -> onboarding_complete -> pehla expense_add (24 ghante me). Kis onboarding step pe log chhod rahe hain wo seedha dikhega.
- **AI monetization:** no_coins_shown -> reward_ad_request -> reward_ad_completed -> coins_credited -> coins_spent. Ye batayega coin model kaam kar raha hai ya log app chhod rahe hain.
- **Siri adoption:** siri_toggle(on) -> siri_how_to_play -> pehla expense_add(source=siri).
- **Retention:** GA4 ka built-in cohort report, legacy_user aur trial_state se split.

> **Privacy (zaroori):** Play Console ka Data safety form aur App Store privacy label update karna hoga ("App interactions", "Device IDs"). Privacy policy me analytics + ads ka zikr. UMP consent jo old app me hai wahi Analytics consent mode ko bhi feed karega.

4. Feature 2 – Siri: har baar lamba sentence na bolna pade

4.1 Pehle sach: iOS kya allow karta hai, kya nahi

Chahiye	Possible?	Wajah
Sirf "150 coffee" bolo, bina "Hey Siri" aur bina app naam	**Nahi** (Siri ke through)	Apple ka rule: har App Shortcut phrase me app ka naam hona zaroori hai. Free-text (String) parameter phrase ke andar allowed nahi - sirf AppEnum/AppEntity. Ye kisi bhi app ke liye nahi badalta.
"Hey Siri, FixMoney" jaisa chhota phrase	**Haan**	Phrase chhota ho sakta hai, bas app ka naam ho. iOS 17+ supported devices pe "Hey" bhi optional hai: "Siri, FixMoney".
Bina Siri ke ek tap/gesture se seedha bolna	**Haan**	Back Tap (phone ke peeche 2 baar tap), Action Button, Control Center button, Lock Screen widget - ye sab hamara shortcut chala sakte hain, koi phrase nahi bolna.
Ek hi saans me "FixMoney 150 coffee"	**Nahi**	Upar wala rule. Siri pehle intent pehchanta hai, phir amount poochta hai (2 step).

4.2 Solution: 3 layer – user jo pasand kare

Layer	User kya karega	Hum kya banayenge	Cost / requirement
A. Chhota Siri phrase	"Hey Siri, FixMoney kharcha" -> Siri: "Kitna aur kis par?" -> "150 coffee"	AppShortcut phrases chhote + Hinglish/English variants: "FixMoney expense", "FixMoney kharcha", "Spend in FixMoney" (code me naam ki jagah \(.applicationName)). Dialog bhi chhota: "Kitna, kis par?"	Free. Real device pe test chahiye (simulator me Siri nahi).

Layer	User kya karega	Hum kya banayenge	Cost / requirement
B. Back Tap / Action Button (**recommended**)	Phone ke peeche 2 tap -> seedha mic sunta hai -> "150 coffee" -> saved. **Na "Hey Siri", na app naam.**	Ek ready-made Shortcut (iCloud link): "Dictate Text" -> "FixMoney: Add expense from text". User ek baar add karke Back Tap / Action Button pe assign kare. How-to video me ye step dikhaenge.	Free. Back Tap: iPhone 8+ / iOS 14+. Action Button: iPhone 15 Pro aur naye.
C. Control Center / Lock Screen button	Control Center kholo -> FixMoney mic button -> bolo	WidgetKit extension + ControlWidget (iOS 18+) jo wahi intent chalata hai	Free, lekin naya Xcode target - Phase 3.

Teeno layer ek hi Kotlin path pe khatam hote hain: `VoiceQuickAddBridge.addFromSpeech(text)` -> Groq parse (Worker se, coin check ke saath) -> category match -> Room save. Naya intent sirf ek aur entry point hai.

4.3 Intent me badlav

- **AddExpenseFromTextIntent** (naya): parameter text, openAppWhenRun = false. Shortcut app / Back Tap isse chalate hain (Dictate Text ka output seedha isme).
- Dono intents **pehle toggle check karenge**: Siri Quick-Add Settings me off hai to Siri bolega: "FixMoney Settings me Siri Quick-Add on karein." (App Shortcuts runtime pe unregister nahi hote, isliye gate intent ke andar.)
- **Coins khatam** ho to Siri bolega: "Coins khatam. FixMoney kholein aur ek video dekhein - 5 coins milenge." Ad Siri ke andar nahi chal sakta (koi UI nahi) - isliye intent ek OpenWallet option deta hai jo app ko Wallet screen pe kholta hai.
- Response chhota: "Rs 150 - Food me save." (abhi lamba message hai).

5. Feature 3 – Settings toggle, permission aur "How to use" video

5.1 Settings row (Android jaisa)

Settings -> AI section me ek row: **"Siri Quick-Add"** (iOS) / existing **"Floating Voice Quick-Add"** (Android). Dono platform pe same flow, sirf permission alag. Feature flag `siri_quick_add` naya add hoga (fail-open, baaki flags jaisa).

5.2 Flow (state machine)

Step	Kya hota hai
1. User toggle ON karta hai	Pref save: siri_enabled = true. Analytics: siri_toggle(on).
2. Permission maango	iOS: `INPreferences.requestSiriAuthorization` -> system dialog "Allow FixMoney to use Siri?". (Android: overlay + mic, jo already bana hai.)
3a. Allowed + **pehli baar** enable	Turant ek card/sheet: **"Kaise use karein"** (play button ke saath). Pref: how_to_seen = true tab set hoga jab video khula.
3b. Allowed + pehle dekh chuka	Kuch nahi dikhana. Sirf toggle ON.
3c. Denied	Toggle wapas OFF, row ke neeche: "Siri permission band hai - Settings kholein" + How-to link (yahan dikhana allowed hai, kyunki permission disabled hai).
4. Baad me app khulne pe	Har baar Settings screen pe permission re-check (Android ka refresh() jaisa). Agar user ne iOS Settings se Siri band kar diya: toggle OFF + "How to use" dobara dikhao. Warna kabhi nahi.

> **Note (honest):** "Hey Siri" pehli baar chalane pe jo confirm Siri khud poochta hai, wo Apple system UI hai - app usse pehle se approve nahi kar sakta. requestSiriAuthorization usko app ke andar le aata hai, lekin ye kis iOS version pe Siri ka first-run prompt poori tarah hata deta hai, ye **real iPhone pe verify karna padega**. Isme Siri capability lagti hai jo **paid Apple Developer account** maangti hai.

5.3 Video: YouTube, bina YouTube app khole

Platform	Approach	Kyun
Android	android-youtube-player library (Apache-2.0, official IFrame API wrap karti hai) - AndroidView me	Free, app ke andar play, YouTube ToS ke andar (official embedded player).
iOS	WKWebView: youtube-nocookie.com/embed/<id>?playsinline=1, allowsInlineMediaPlayback = true - UIKitView me	playsinline ke bina iOS full-screen player kholta hai; nocookie se tracking kam.
Common	expect/actual composable `HowToVideo(videoId)` ek bottom sheet me	Screen code ek hi rahega.

- Video ID **Firestore** doc `app_config/app` me: `howto_siri_video_id`, `howto_voice_video_id`. Naya video upload karo, ID badlo - app update nahi chahiye.
- YouTube pe video **Unlisted** chal jayega, bas "Allow embedding" ON ho.
- Internet nahi hai to: "Video ke liye internet chahiye" + 3 line ka text guide fallback.
- Firebase Storage me MP4 rakhna avoid kiya: naye projects me Storage ke liye Blaze (card) plan lagta hai, YouTube free hai aur bandwidth bhi unka.

6. Feature 4 – AI Coins: rules

Rule	Value (default, server pe badal sakte)
Free trial	**5 din** - pehli baar server pe register hone se. Trial me sab AI features bina coin (fair-use cap 30 uses/din, Groq quota bachane ke liye).
Trial ke baad	1 coin = 1 AI use. Coins 0 to AI feature pe "Video dekho, 5 coins pao" sheet.
Rewarded video	1 poora video = **5 coins**. Max 10 video/din (= 50 coins) - AdMob invalid-traffic se bachne ke liye.
Kis cheez pe coin lagega	Fixoo message: 1 · Voice add (bubble ya Siri): 1 · Receipt scan: 1 (recommended - ye bhi Groq kharch karta hai) · AI Insight: section 7.7
Refund	AI call fail hui (Groq + fallback dono down, parse fail) -> coin wapas. User ko kharab result ka paisa nahi.
Free hamesha	Manual expense, budget, analytics, export, SIP - koi coin nahi.
Old users (update)	Unka trial bhi pehli KMP launch se 5 din. Optional welcome bonus (jaise 10 coins) - config se.

7. Coins – technical design

7.1 Identity: bina login, bina Firebase Auth

User se **koi login, email, phone ya Google sign-in nahi** maanga jayega, aur Firebase Auth bhi use nahi hoga. Pehchaan sirf phone ki **device key** (7.2) se hogi:

- Pehli launch pe app chupke se `POST /v1/register {deviceKey}` bhejta hai.
- Worker us device ka account dhundhta hai ya naya banata hai, aur ek **device token** lautata hai: `accountId.issuedAt.HMAC-SHA256(secret)`. Secret sirf Worker me (wrangler secret), jaise Groq key.
- App token Settings (Android) / Keychain (iOS) me rakhta hai, aur har request me `Authorization: Bearer <deviceToken>` bhejta hai.

- Worker HMAC check karta hai (microseconds, koi network call nahi, Google keys fetch nahi). Token nakli ya badla hua = 401.
- Token kho gaya (reinstall)? App phir se register karta hai, same device key -> **same account**, coins aur trial wahi.

> **Trade-off (jaan bujh kar):** login nahi hai, isliye **naya phone ya factory reset = naya account** - purane coins transfer nahi honge. Coins ki koi cash value nahi aur free me kamaaye jaate hain, isliye ye theek hai. Wallet screen pe ek line: "Coins is phone se jude hain." Firebase Auth hatane se ek dependency aur ek network hop bhi kam hua.

7.2 Device key: reinstall karke trial reset na ho

Platform	Key	Reinstall pe?
Android	SHA-256(ANDROID_ID + app salt)	Same rehta hai (Android 8+ pe ANDROID_ID signing key + device ke hisaab se fix). Factory reset pe badalta hai.
iOS	Keychain me rakha random UUID	Keychain items app delete ke baad bhi bachte hain - reinstall pe same.

Trial aur wallet **device key** se bandhe hain. Reinstall pe key same rehti hai, isliye coins aur trial dono wahi.

7.3 Ledger storage: Cloudflare D1 (recommended) vs Firestore

	Cloudflare D1	Firestore (Spark)
Free writes/day	**100,000** rows	20,000 docs
Free reads/day	5,000,000 rows	50,000 docs
Worker se access	Native binding, ek line - koi key nahi	REST + service-account JWT sign karna padega Worker me
Atomic "coins >= 1 ho to minus 1"	Ek SQL UPDATE ... WHERE coins >= 1	Transaction (read + write = 2 ops)
Latency	Worker ke saath same network	Extra hop Google tak
Kab tak free (approx)	~15-20k DAU (5 AI use/din)	~2-3k DAU

> **Recommendation:** Firebase ko **analytics + config** ke liye use karo (jo wo best karta hai), aur coin ledger **D1** me - kyunki coin check usi Worker me hota hai jo AI call karta hai. Agar aap Firestore pe hi rehna chahte ho to design same rahega, sirf storage adapter badlega; tab 20k writes/din ki limit yaad rakhni hogi.

7.4 Data model (D1)

```
accounts (account_id PK, device_key UNIQUE, created_at, trial_ends_at, platform)
wallets (account_id PK, coins INT CHECK(coins >= 0), updated_at)
ledger (id PK, account_id, delta INT, reason, feature, ref UNIQUE, created_at)
        -- reason: ad_reward | ai_use | refund | bonus
        -- ref: AdMob transaction_id ya app ka request_id -> duplicate kabhi count nahi hoga
ad_daily (account_id, day, count, PRIMARY KEY(account_id, day))
config (key PK, value)   -- trial_days=5, reward_per_ad=5, daily_ad_cap=10, cost_fixoo=1 ...
```

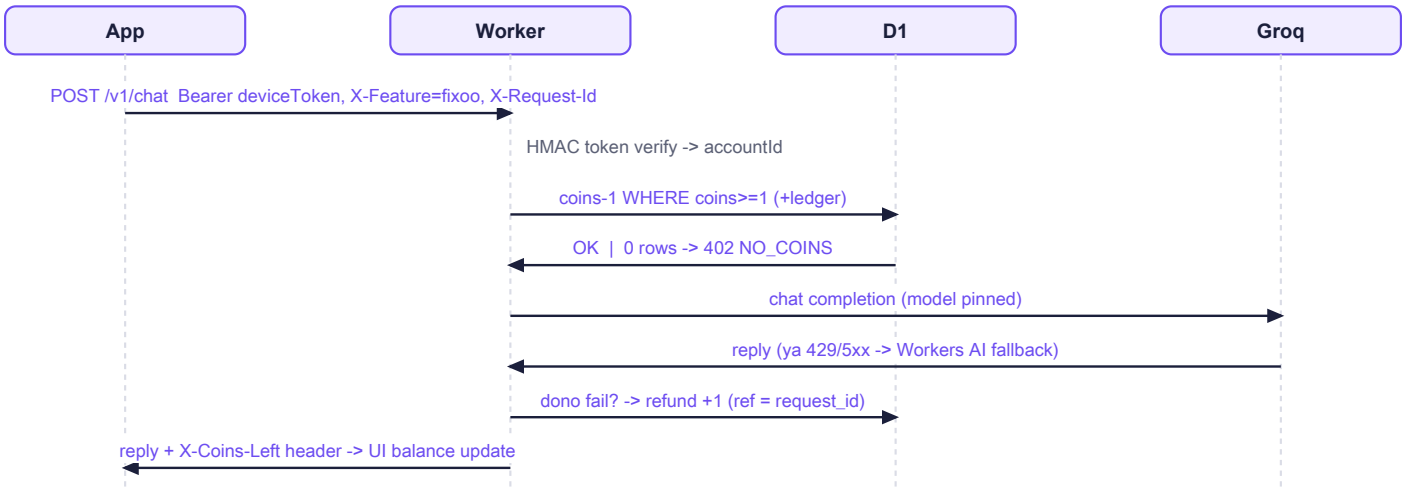
7.5 API (Worker)

Endpoint	Kaun call karta hai	Kya karta hai
POST /v1/register	App (launch pe, ek baar)	device_key se account dhundho/banao -> signed deviceToken + {coins, trialEndsAt, costs}

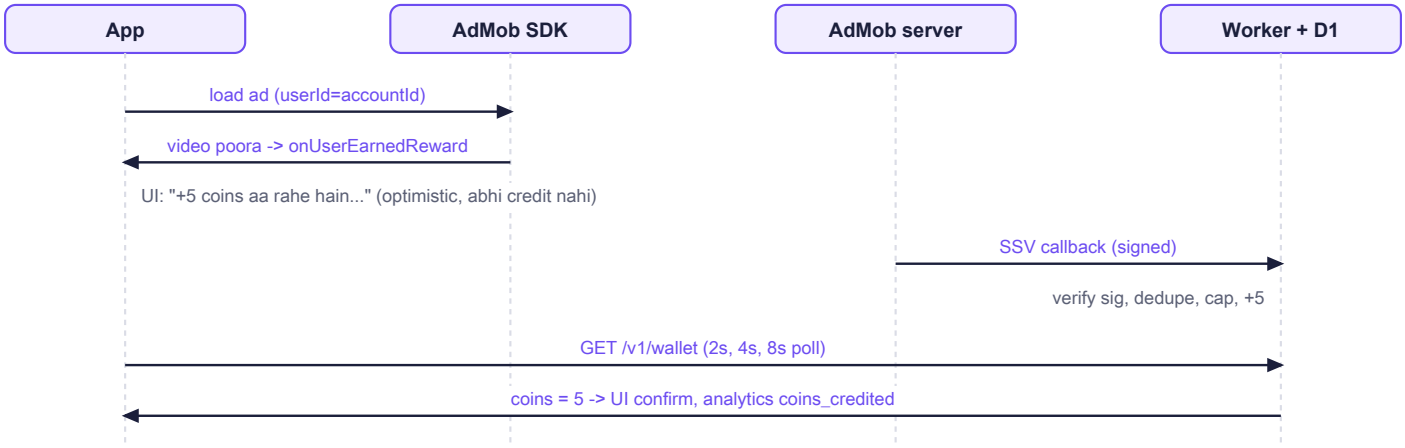
Endpoint	Kaun call karta hai	Kya karta hai
GET /v1/wallet	App (Wallet screen, ad ke baad)	{coins, trialEndsAt, adsLeftToday, costs}
POST /v1/chat	Fixoo, voice, receipt (existing)	Headers: X-Feature, X-Request-Id. Trial active? -> fair-use check. Warna atomic debit. Phir Groq/fallback. Fail -> refund. Coins nahi -> **402 NO_COINS**.
POST /v1/spend	Non-LLM paid features (insight agar paid ho)	Sirf debit, idempotent by request_id
GET /v1/admob/ssv	**AdMob ka server** (app nahi)	Signature verify (Google keys) -> transaction_id unique insert -> daily cap -> +5 coins

7.6 Flows

(a) AI feature use (trial ke baad)



(b) Video dekh kar coins kamana



- App kabhi khud +5 nahi deta. SSV aksar kuch second me aata hai; agar 30s me na aaye to "Coins thodi der me jud jayenge" - transaction_id ki wajah se baad me pahuncha callback bhi sahi credit hoga.
- AdMob console me Rewarded unit pe **Server-side verification** ON karke callback URL = Worker ka /v1/admob/ssv.
- Old app ki seekh port karni hai: UMP consent ad request se pehle, test device IDs debug me, aur apne phone pe real ad kabhi click nahi (account suspend ho sakta hai).

7.7 AI Insight ka sawaal

Insight card abhi **local calculation** hai, AI call nahi - hamara 0 kharch. Do option:

Option	Kaise	Fayda / nuksaan
A. Free rakho (recommended)	Insight hamesha dikhega, coin nahi	Ye Home pe "aha" moment hai jo user ko AI features try karne pe laata hai. Kharch zero, toh lock karne ka sirf nuksaan.
B. Din me 1 coin	Card blur + "1 coin me unlock (aaj ke liye)" -> /v1/spend	Thodi zyada ad views, lekin cheez jo free ho sakti thi uske liye user chidh sakta hai.

Dono design support karta hai - `cost_insight` config me 0 ya 1. Aapko decide karna hai (section 12).

7.8 App side pieces

Piece	Kahan	Kaam
DeviceIdentity	shared/identity	Device key banana, /v1/register, token store (koi login UI nahi)
WalletRepository	shared/coins	StateFlow<Wallet> (coins, trialEndsAt, adsLeft); /v1/register + /v1/wallet
AiProxy (badlav)	shared/ai	deviceToken + X-Feature + X-Request-Id headers; 402 -> NoCoins result (har caller handle karega)
RewardedAdController	composeApp expect/actual	Android: play-services-ads; iOS: Google Mobile Ads Swift package + bridge (Firebase wala pattern)
ConsentManager	composeApp androidMain/iosMain	UMP - old AdManager se port
WalletScreen + NoCoinsSheet	ui/coins	Balance, trial ke din baaki, "Video dekho +5", aaj kitne video baaki, history
Coin chip	Home header + Fixoo header	Balance hamesha dikhe, tap -> Wallet
Flags	feature_flags/app	coins_enabled (kill switch), rewarded_ads

7.9 Cheating aur edge cases

Risk	Bachav	Kitna bacha
App patch karke coin badhana	Balance sirf server pe	Poora
Nakli "ad dekha" request	Sirf signed SSV se credit, signature verify	Poora
Same ad callback 2 baar	transaction_id UNIQUE	Poora
Retry pe 2 baar coin katna	X-Request-Id UNIQUE ledger me	Poora
Reinstall karke naya trial	Device key (ANDROID_ID hash / Keychain)	Zyada tar; factory reset pe naya
Script se nakli device keys	Naye account per-IP cap (KV), baad me Play Integrity (free 10k/din)	Partial - lekin nuksaan sirf free AI quota, paisa nahi
Groq ka free quota khatam (1,000 req/din poore org ka)	Workers AI fallback, trial fair-use cap, ad cap	Section 8 dekhein
Offline	AI vaise bhi internet maangta hai; balance last known dikhao	-
Ghadi peeche karke trial badhana	Trial end time server ka	Poora

8. Free resources aur scale ki seemayein

Service	Free limit	Hamara use	Pehli deewar kab
Firebase Analytics	Unlimited events (500 event names, 25 user props)	Saare events	Kabhi nahi
Firestore	50k reads / 20k writes per din	Sirf config (cached 6-24h)	Bahut door
Cloudflare Workers	100k requests/din, 10ms CPU	AI + wallet + SSV	~15k DAU; phir $5/mahina
Cloudflare D1	5M reads, 100k writes/din, 5 GB	Ledger	~15-20k DAU
YouTube embed	Free	How-to video	Kabhi nahi
AdMob	Free (paise dene wala hai)	Rewarded	-
Groq free	**~1,000 requests/din poore org ke liye**	Har AI use	**Sabse pehle: ~200 active AI users/din**
Workers AI	10,000 Neurons/din	Groq fallback	Groq ke saath thoda extra

> **Asli bottleneck AI quota hai, coins nahi.** Free me poore app ke liye roz ~1,000 AI calls hain. Coin system ka asli fayda yahi hai: jo user zyada AI use karta hai wo ad dekhta hai, aur ad ka paisa paid Groq ka bill bharta hai. Isliye Groq ko **paid tier** pe le jaana jab bhi daily calls ~800 cross karein - Worker me sirf key badalni hai, app update nahi.

8.1 Unit economics (estimate - asli numbers verify karne hain)

Cheez	Andaza	Note
1 AI call ka token	~1,500 input + 300 output	Fixoo prompt me DATA summary hota hai
gpt-oss-20b paid rate	~$0.075 / M input, ~$0.30 / M output	**Assumption** - Groq pricing page pe confirm karein
1 AI call ka kharch	~$0.0002	1,500 x 0.075/M + 300 x 0.30/M
5 coins (1 video) ka kharch	~$0.001	5 AI calls
1 rewarded view ki kamai	eCPM / 1000	India me eCPM bahut vary karta hai - **pehle 2 hafte ka AdMob data dekh kar** reward_per_ad tune karein

Formula: **profit per video = (eCPM / 1000) - (reward_per_ad x cost per AI call)**. Kyunki reward_per_ad aur costs server config me hain, ise bina app update tune kar sakte hain.

9. Policy aur compliance checklist

- **AdMob rewarded policy:** ad hamesha user ke tap se shuru ho, reward pehle se clearly likha ho ("5 coins"), ad pe click karne ka reward nahi. Core expense tracking ko ad ke peeche nahi rakha - sahi.
- **Play Data safety + App Store privacy label:** analytics, device ID, ads ke liye update.
- **Privacy policy** (old app ka docs/PRIVACY.md): Firebase Analytics, AdMob, device key (hashed), Worker me kya store hota hai (account id, coin ledger - koi expense data nahi).
- **Consent:** UMP form EEA/UK me ads se pehle; India-first app hai lekin form sab jagah safe hai.
- **iOS ads:** App Tracking Transparency prompt (optional; na ho to non-personalized ads, kam eCPM), SKAdNetwork IDs Info.plist me.
- **Misrepresentation:** coins ka koi cash value nahi, khareed nahi sakte - UI me "coins sirf AI features ke liye" likhna. (In-app purchase se coins bechna Apple/Google billing maangega - ye scope me nahi.)

10. Blockers jo pehle se pata hone chahiye

Blocker	Kya rukta hai	Rasta
Paid Apple Developer account nahi ($99/saal)	Siri capability (requestSiriAuthorization), real iPhone pe Siri test, iOS AdMob live, App Store release	iOS ka code likh sakte hain aur simulator pe compile/run; Siri + ads ka asli test account ke baad
Simulator me Siri/camera nahi, tap nahi kar sakte	Siri flow ka machine test	Android pehle poora verify (OnePlus Nord), iOS device pe baad me
Android 12 test device	Android 14+ specific behaviour (jaise microphone FGS) yahan nahi dikhte	Ek Android 14+ device ya Firebase Test Lab (free quota)
Groq free quota	Launch ke baad AI capacity	Paid Groq jab daily calls ~800 cross karein

11. Implementation phases

Phase	Kya	Kitna (approx)	Verify kaise
1. Analytics	Bridge + Firebase setup dono platform, saare journey events, DebugView check	2-3 din	DebugView me har journey stage ka event, OnePlus pe
2. Coin backend	D1 schema, /register, /wallet, /chat gate + refund, /admob/ssv, config	3-4 din	curl se: debit, 402, refund, duplicate SSV, daily cap
3. Coin app (Android)	Device identity, Wallet, Rewarded ad + UMP, NoCoins sheet, coin chip, analytics	3-4 din	Test ad unit se poora loop real device pe
4. Siri + toggle + video	Settings row, permission flow, HowToVideo (dono), naya text intent, chhote phrases, Shortcut link, gate + no-coins reply	3 din	Android toggle/video device pe; iOS simulator compile + render
5. iOS ads + Siri device test	Google Mobile Ads iOS, ATT, real iPhone pe Siri	2 din	Apple account ke baad
6. Control Center button (optional)	WidgetKit + ControlWidget	1-2 din	iOS 18 device

Order ki wajah: Analytics pehle, taaki coins launch hone se **pehle** ka baseline data mile - warna pata nahi chalega coins ne retention badhaya ya giraya.

12. Aapke decisions chahiye

#	Sawaal	Meri recommendation
1	Coin ledger: D1 ya Firestore?	D1 (5x zyada free writes, Worker ke andar hi). Firebase sirf analytics + config ke liye; login nahi.
2	AI Insight pe coin lagega?	Nahi - free rakho, ye kharch nahi karta aur AI ka trailer hai.
3	Receipt scan bhi coin pe?	Haan, 1 coin - ye bhi Groq call karta hai.
4	Daily video limit	10 video / din
5	Old users ko welcome bonus?	10 coins, taaki update ke baad kuch bura na lage
6	Siri layer B (Back Tap Shortcut) pehle?	Haan - ye hi asli "bina sentence bole" experience deta hai, bina naye Xcode target ke
7	Banner / interstitial bhi KMP me laane hain (old app jaise)?	Abhi nahi - rewarded pe focus; banners baad me flag ke peeche

Approval ke baad Phase 1 se shuru. Har phase ke baad CLAUDE.md update hoga (decisions + kya verify hua).